

Heuristics for finding IF

Option 1. Assume $\mu(x, y) = x^m y^n$. Solve for m and n from exactness condition.

See <https://www.mathsisfun.com/calculus/differential-equations-exact-factors.html>.

Option 2. Assume $\mu(x, y) = \mu(x)$. Exactness condition must be variable-separable in μ and x . Solve for $\mu(y)$.

Option 3. Assume $\mu(x, y) = \mu(y)$. Exactness condition must be variable-separable in μ and y . Solve for $\mu(y)$.

Find an IF, then solve.

$$\cos x \, dx + \left(1 + \frac{2}{y}\right) \sin x \, dy = 0$$

$$\mu = e^y y^2$$
$$y^2 e^y \sin x = C$$

Find an IF, then solve.

$$y(x + y + 1) dx + (x + 2y) dy = 0$$

$$\mu = e^x$$
$$e^x y(x + y) = C$$

Verify the IF, then solve.

$$(x^2 + 2xy - y^2) dx + (y^2 + 2xy - x^2) dy = 0$$

$$\mu = (x + y)^{-2}$$

$$x^2 + xy + 2y^2 = C(x + y)$$

Make it exact.

$$M(x, y) dx + (xe^{xy} + 2xy + x^{-1}) dy = 0$$

$$M(x, y) = ye^{xy} + y^2 - yx^{-2} + K(x)$$